

$$\rightarrow d = 2 \frac{gm}{ml}$$

$$m = M$$

$$\frac{n_{st}}{w_{solu}} = \frac{n_{st}}{V}$$

$$w_{solu} = V_{soln}$$

$$d = \frac{w_{soln}}{V_{soln}}$$

$$\neq \frac{w_{st} + w_{solu}}{w_{solu}}$$

$$2 w_{solu} = w_{st} + w_{solu}$$

$$w_{solu} = w_{st}$$

$$\frac{x_{st}}{x_{solu}} = \frac{n_{st}}{n_{solu}}$$

$$\frac{1}{g} = \frac{n_{st}}{\frac{w_{st}}{M \cdot w_{solu}}}$$

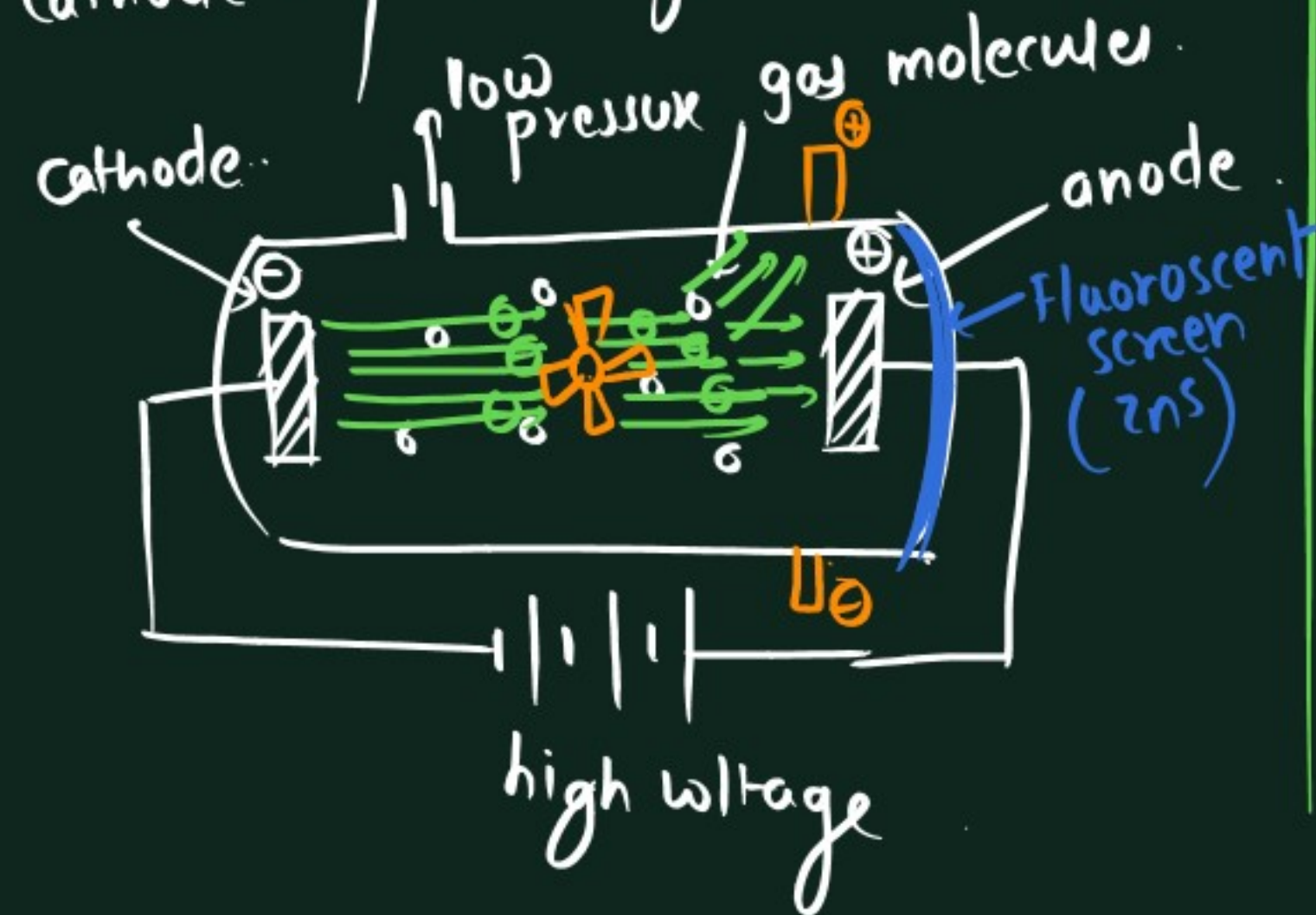
$$\frac{1}{g} = \frac{w_{st}}{w_{solu}} \times \frac{M \cdot w_{solu}}{M \cdot w_{st}}$$

Atomic Structure

* Discovery of subatomic particles

* Discovery of e^-

Cathode ray discharge tube



- ① some rays travel from cathode to anode $\rightarrow$ cathode rays
- ② Cathode rays travel in straight line
- ③ Cathode rays are -vely charged.
- ④ Cathode rays gets deflected under influence of electric & mag. Field.
- ⑤ Cathode rays can rotate paddle wheel. Shows that it has some mass.
- ⑥ Cathode rays ~~can~~ when strike on ZnS screen it glows

* J.J Thomson → determine specific charge of e^- .
→ charge of 1 kg of e^-
specific charge = $1.75 \times 10^{11} \frac{\text{Coulomb}}{\text{kg}}$

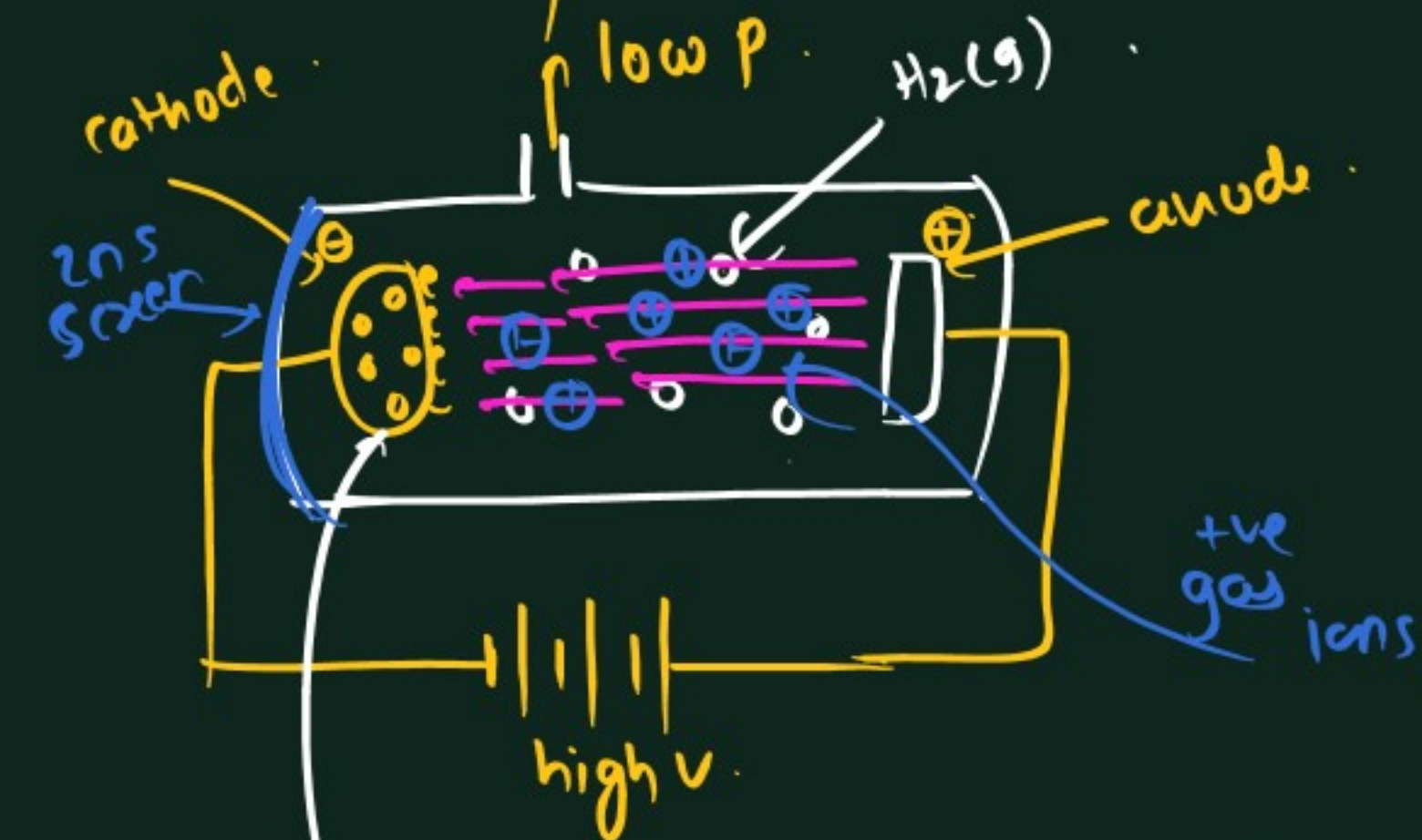
* Milliken's oil drop method →
→ oil drops passed thr' discharge tube
& collected oil drop does have some charge
→ charge of $1e^- = -1.602 \times 10^{-19} \text{ C}$
-ve sign shows that e^- is -vely charged

$$\begin{aligned}\text{Mass of } e^- &= \frac{1.6 \times 10^{-19}}{1.75 \times 10^{11}} \\ &= \underline{\underline{9.1 \times 10^{-31} \text{ kg}}} \\ &= \underline{\underline{9.1 \times 10^{-28} \text{ gm}}}\end{aligned}$$

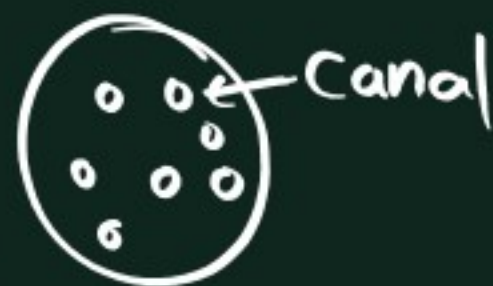
Note → Cathode rays expt.
doesn't depend on
nature of gas used

Discovery of proton →

→ Goldstein
→ Anode ray expt.



Cathode → Perforated cathode
→ contains small holes



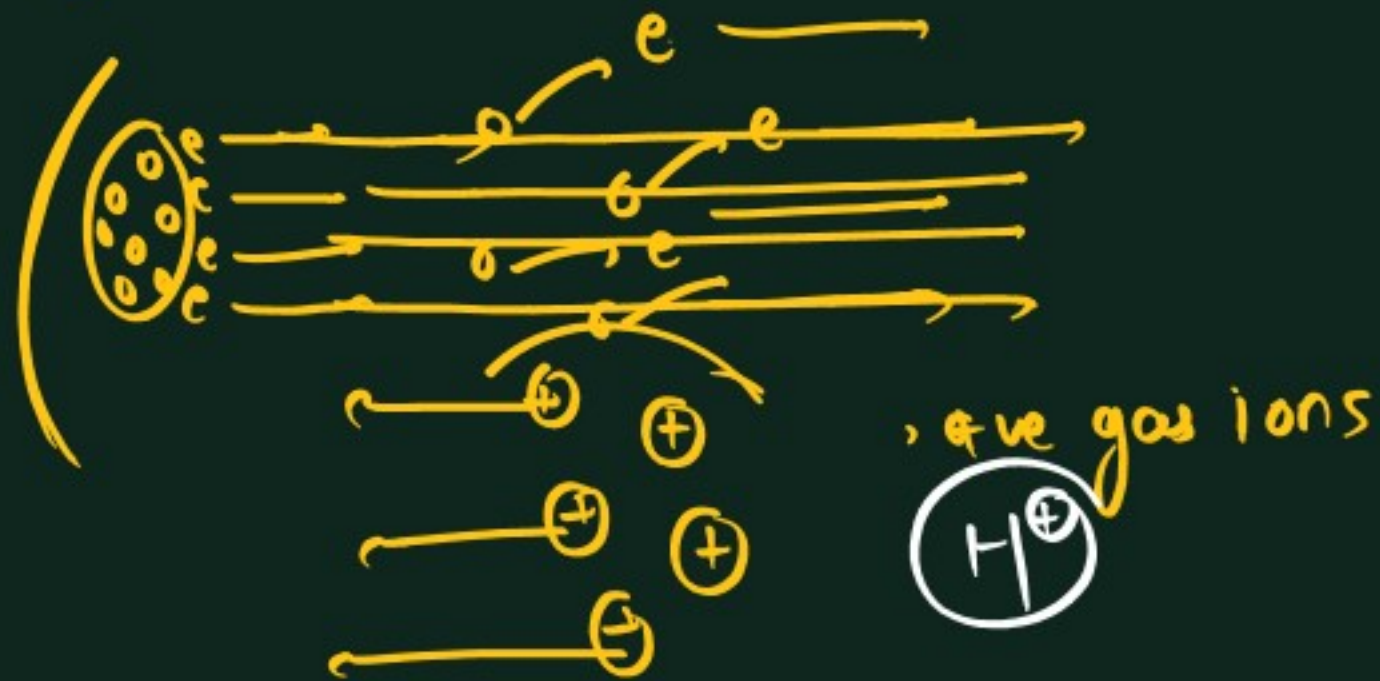
- 1) Some rays travel from anode to cathode → anode rays or canal rays
- 2) They are +vely charged.
- 3) They are deflected under influence of electric & magnetic field.
- 4) They can rotate paddle wheel.
- 5) They can glow ZnS screen.
- 6) Anode rays named as proton.

$$\text{charge} = +1.602 \times 10^{-19} \text{ C}$$

$$\text{mass} = 1.67 \times 10^{-27} \text{ kg} = 1.67 \times 10^{-24} \text{ gm}$$

$$\text{specific charge} = 9.6 \times 10^7 \text{ C/kg}$$

Δ No Anode ray expt depends nature of gas used.

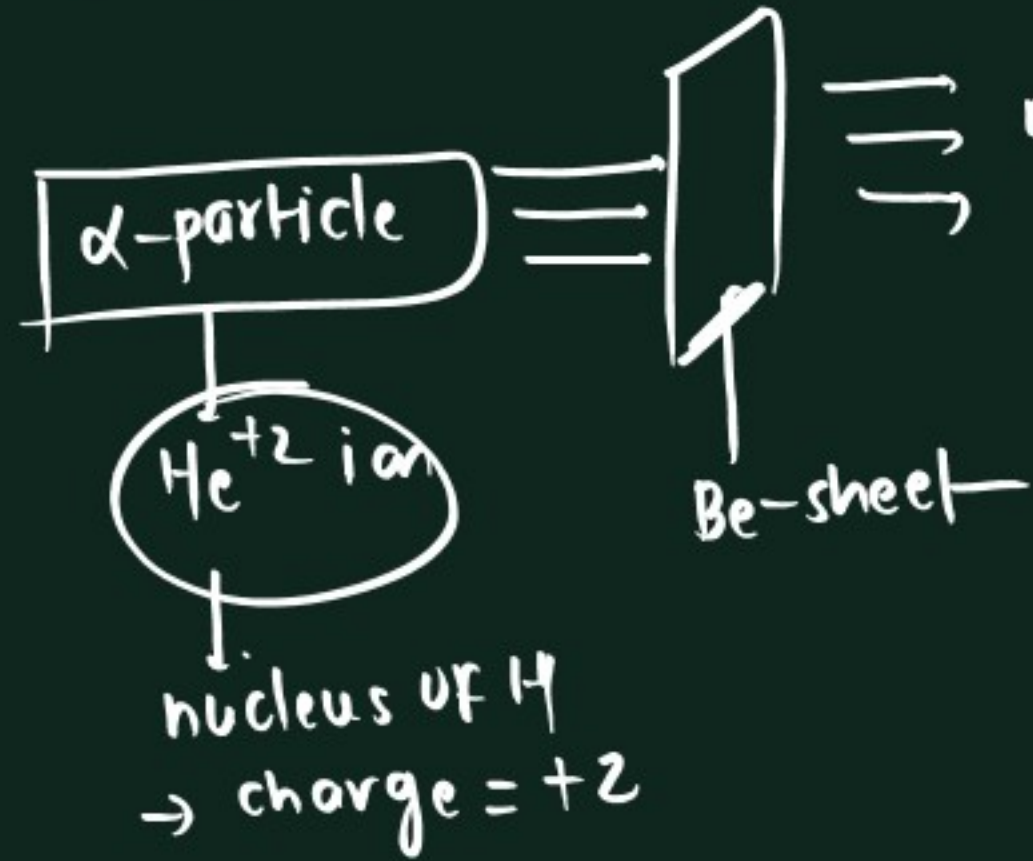


e.g $H_2(g)$ is used.

	e	p	n
H_2	1	1	0
H^+	0	✓1	0

	e	p	n
He	2	2	2
He^{+2}	0	2	2

Discovery of neutron →
→ Chadwick.



neutral particle → neutrons.
→ charge = 0
mass = 1.67×10^{-27} kg.



* Properties	proton	electron	neutron
1) mass	$1.67 \times 10^{-27} \text{ kg}$	$9.1 \times 10^{-31} \text{ kg}$	$1.67 \times 10^{-27} \text{ kg}$
2) charge	$+1.602 \times 10^{-19} \text{ C}$	$-1.602 \times 10^{-19} \text{ C}$	—
3) Relative charge	+1	-1	—
4) Relative mass	1	$\frac{9.1 \times 10^{-31}}{1.67 \times 10^{-27}} = \frac{1}{1837} \approx \frac{1}{1840}$	1



4) $\frac{\text{charge}}{\text{mass}} \cdot \left(\frac{e}{m}\right)$ ratios

for d, p, e, n .

$$d = \frac{2}{4} = \frac{1}{2}$$

$$p = \frac{1}{1}$$

$$e = \frac{1}{\frac{1}{1840}} = 1840$$

$$n = \frac{0}{1}$$

$$n < d < p < e$$

2) charge of $1e^- = 1.6 \times 10^{-19} \text{ C}$
 charge of 1 mole = $1.6 \times 10^{-19} \times N_A$
 $= 1.6 \times 10^{-19} \times 6 \times 10^{23}$
 $= \underline{96500 \text{ C}}$
 $= 1 \text{ faraday (1F)}$

3) calculate charge of 2 moles of α -particle.
 $\rightarrow$ charge of 1 mole of proton = 1F
 charge of 4 mole of protons = 4F

4) charge of oil drop = 6.4×10^{-18} $\rightarrow x$
 $1.6 \times 10^{-19} \rightarrow 1e^-$

$$x = \frac{6.4 \times 10^{-18}}{1.6 \times 10^{-19}}$$

$$= \underline{40}$$

A) $x/2$

B) $2x$

C) $x/4$

☒ D) x

a) gets doubled

b) gets halved

c) Remain same

☒ d) Reduced by 25%

% Reduction = $\frac{\text{change}}{\text{Initial}} \times 100$

$$= \frac{64}{24} \times 100 = 25\%$$

$$mg = 12P + 12n$$

$$= 24$$

$$mg = 12P + 6$$

$$= \underline{18}$$

Atomic no. (Z) = no. of proton
Mass no. (A) = no. of p + n

Iso-series →

1) Isotope → all atoms of same element
→ same atomic no.
→ diff. mass no.
→ similar properties.

e.g.) ${}^1_1\text{H}^1$, ${}^2_1\text{H}^2$, ${}^3_1\text{H}^3$
protium Deuterium Tritium

2) ${}^{35}_{17}\text{Cl}$ / ${}^{37}_{17}\text{Cl}$

3) ${}^{12}_6\text{C}$ / ${}^{14}_6\text{C}$

4) ${}^{14}_7\text{N}$ / ${}^{16}_7\text{N}$

5) ${}^{16}_8\text{O}$ / ${}^{18}_8\text{O}$

2) Isobar →

→ atoms of diff. elements

→ so diff. atomic no.

→ same mass no.

→ Isobars have diff. properties.

e.g.) ${}^{40}_{18}\text{Ar}$ / ${}^{40}_{20}\text{Ca}$

2) ${}^{14}_6\text{C}$ / ${}^{14}_7\text{N}$

3) ${}^{24}_{11}\text{Na}$ / ${}^{24}_{12}\text{Mg}$

3) Isotones →

→ same no. of neutron (A-Z)

e.g.) ${}^3_1\text{H}$ / ${}^4_2\text{He}$

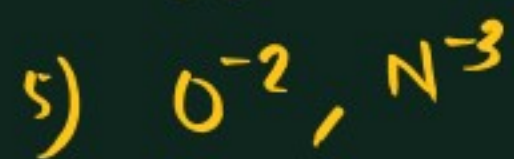
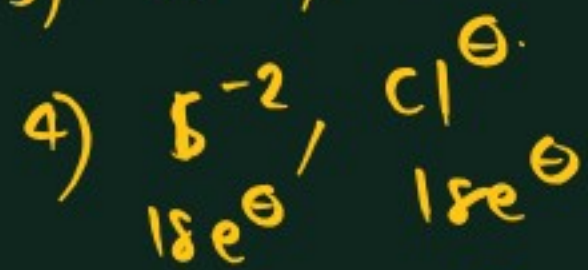
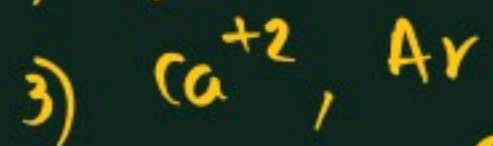
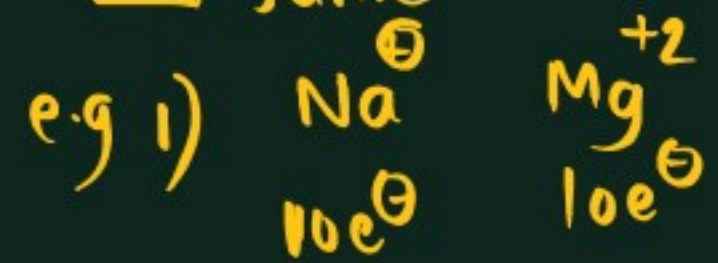
2) ${}^{19}_9\text{F}$ / ${}^{20}_{10}\text{Ne}$

3) ${}^{23}_{11}\text{Na}$ / ${}^{24}_{12}\text{Mg}$

4) ${}^{31}_{15}\text{P}$ / ${}^{32}_{16}\text{S}$

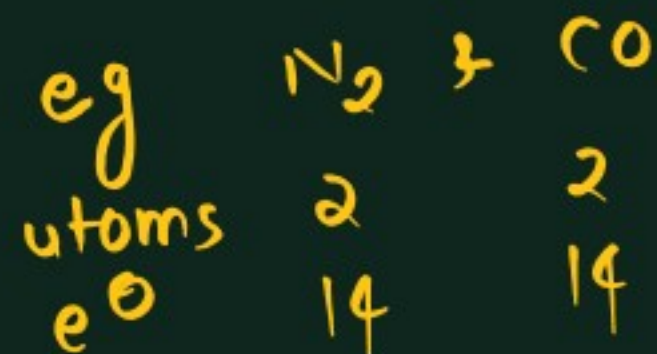
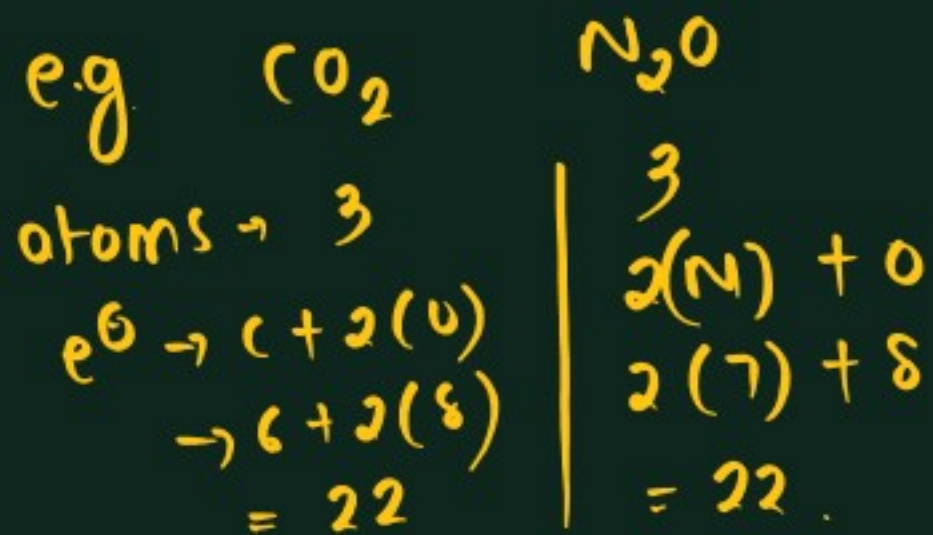
4) Isoelectronic species →

→ Same no. of e^-



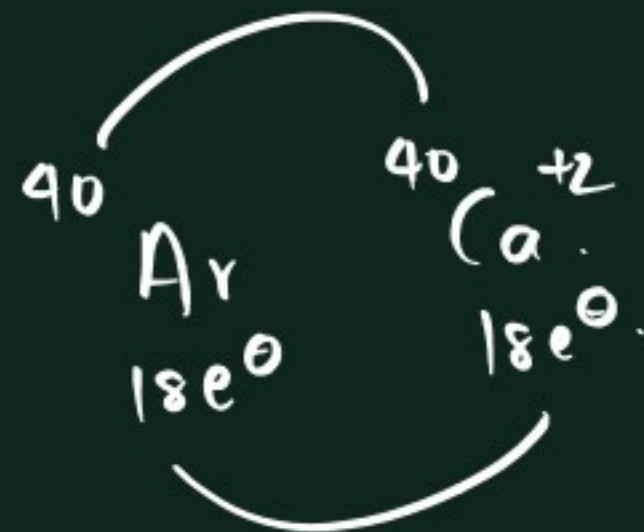
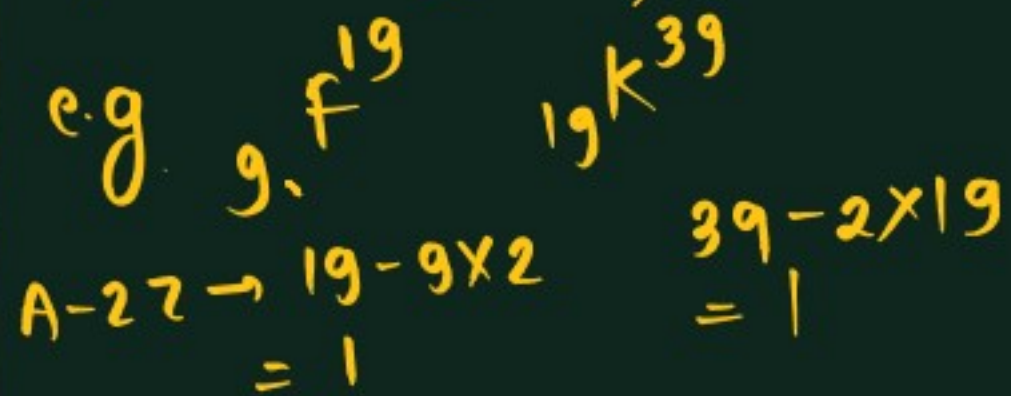
5) Isobesters →

→ Two molecules having same no. of atoms & e^-



6) Iso-diaphers →

→ Same value of neutron excess
 → Same $(A - 2Z)$ or $n - p$



* J.J. Thomson Plum pudding model →

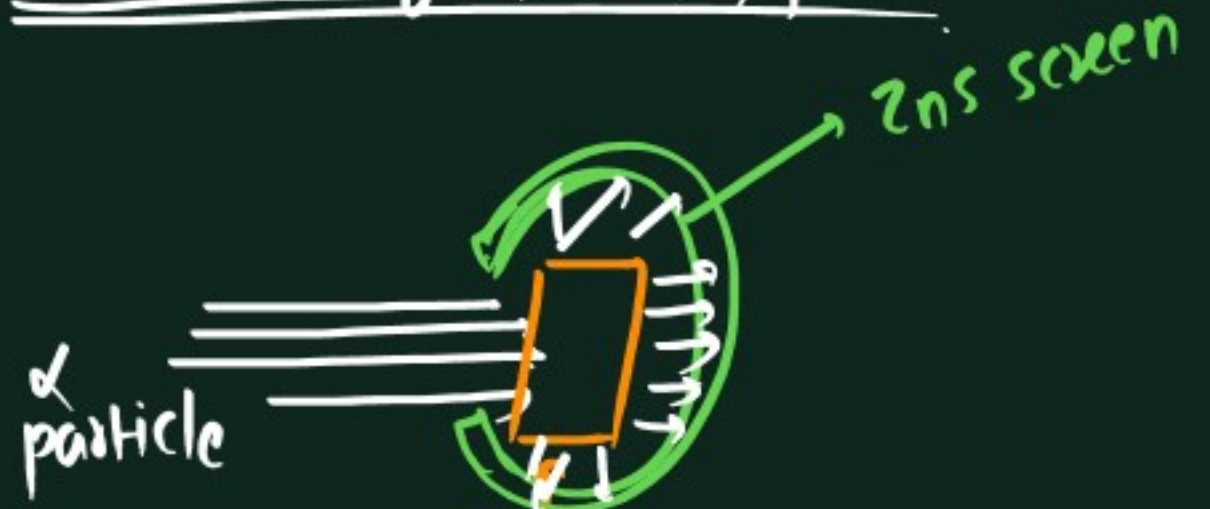
→ embedded



Drawback -

- ① It could not explain stability of an atom
- ② It could not explain Rutherford gold foil expt

* Rutherford gold foil expt →



- gold foil
- Most malleable metal & converted into thin sheet.
 - So that no. of atoms are very less.

Observation →

Conclusion —

①

②



* Model →

Diameter of atom — 10^{-10} m

Diameter of Nu — 10^{-15} m

Radius of Nu = $1.25 \times 10^{-15} A^{1/3} \text{ m}$

$1 \text{ fm} = 10^{-15} \text{ m}$

